

Control work 2, and the quarter review

Prisms and cylinders — area, volume and bounds — in one paper, then the map.

LESSON 31–32

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 2

IGCSE E13 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
Concept map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply every formula of Quarter II in one assessment.
- Choose between curved, total and open-solid surface areas.
- Classify each lost mark and rewrite the solution.
- Build a concept map of mensuration so far.

TEXTBOOK

National	pp. 157–160
Cambridge	Core & Extended, pp. 319–322
Workbook	Control paper G2

01

Explanation

The paper — 30 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Space diagonal of a $4 \times 6 \times 12$ cuboid, and the angle it makes with the base	5	L21–22
2	Total surface area of a regular hexagonal prism, edge 5, height 11	5	L23
3	Volume of a trough with trapezium section 30/50 cm, depth 20 cm, 2.5 m long, in litres	5	L26–27
4	A cylinder holds 800 ml and is 10 cm tall: find r and the curved surface area	5	L28–29
5	A pipe 2 m long, $R = 9$, $r = 7$ cm: metal volume, then mass at 7.8 g/cm^3	5	L28–30
6	A cuboid measures $5 \times 7 \times 9$ cm to the nearest cm. Find the bounds of its volume	5	L24

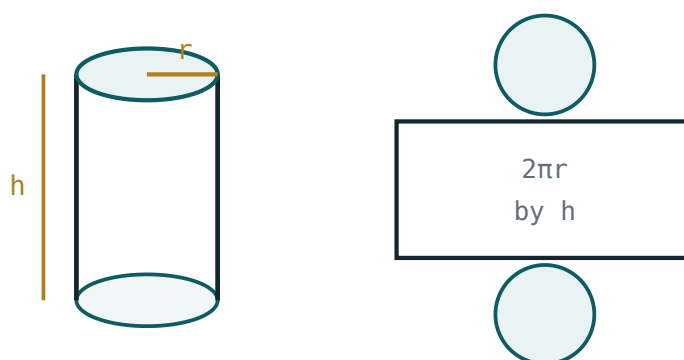
UNITS AND FORMULA CHOICE

Q3 and Q5 each carry a mark for the conversion alone; Q4 carries one for choosing **curved** rather than total. Five of the thirty marks are for reading the question.

The concept map

Six boxes, links as sentences:

- **polyhedral angle** → **only five regular solids** — “the angles must total under 360° ”
- **prism** → **uniform cross-section** — “ $V = Bh$, always”
- **net** → **surface area** — “ $S = 2B + Ph$ ”
- **cylinder** → **the same two formulas** — “with $B = \pi r^2$ and $P = 2\pi r$ ”
- **Cavalieri** → **oblique = right** — “equal slices, equal volume”
- **measurement** → **bounds** — “three lengths, three uncertainties, one wide answer”



$$S = 2\pi r^2 + 2\pi rh - \text{two circles and one rectangle}$$

The picture behind two of the six boxes.

Looking forward

Quarter III is the pyramid, the cone and the sphere — the solids whose volume formulas carry a $\frac{1}{3}$ or a $\frac{4}{3}$. Every one is proved by comparison with the prism and cylinder of this quarter, so those two formulas must be secure.

ONE HABIT TO CARRY FORWARD

Write the base area on its own line before anything else. Every volume in the next quarter starts from B , and every error starts from having computed it wrong.

02 Worked examples

EXAMPLE 1 Model answer, Q3: trough, trapezium 30/50 cm, depth 20 cm, 2.5 m long.

$$1 \quad B = \frac{30 + 50}{2} \times 20 = 800 \text{ cm}^2.$$

2 Length 250 cm.
Convert first.

$$3 \quad V = 800 \times 250 = 200\,000 \text{ cm}^3.$$

$$4 \quad = 200 \text{ litres.}$$

ANSWER

200 litres

EXAMPLE 2 Model answer, Q4: a cylinder holds 800 ml and is 10 cm tall.

$$1 \quad 800 = \pi r^2 \times 10$$

$$2 \quad r^2 = \frac{80}{\pi} \approx 25.46$$

$$3 \quad r \approx 5.05 \text{ cm.}$$

$$4 \quad S_{\text{curved}} = 2\pi rh \approx 317 \text{ cm}^2.$$

Curved, not total.

ANSWER

$r \approx 5.05 \text{ cm}; \approx 317 \text{ cm}^2$

EXAMPLE 3 Model answer, Q6: bounds for a $5 \times 7 \times 9$ cuboid to the nearest cm.

- ① 4.5–5.5, 6.5–7.5, 8.5–9.5.
- ② $V_{\min} = 4.5 \times 6.5 \times 8.5 = 248.625$
- ③ $V_{\max} = 5.5 \times 7.5 \times 9.5 = 391.875$
- ④ Stated volume 315.
The range is nearly 150.

ANSWER248.6 to 391.9 cm³**03** Interactive model

Work Q3 and Q4 on the board, converting units before any arithmetic.

The quarter in ten questions

Plane angles at a vertex sum to:

360°

less than 360°

more

180°

Space diagonal of a $4 \times 6 \times 12$ cuboid:

12

14

22

$\sqrt{196}$

S_{total} of a right prism:

Ph

$2B + Ph$

Bh

$2Ph$

Volume of any prism:

$$Ph$$

$$Bh$$

$$\frac{1}{3}Bh$$

$$2Bh$$

Volume of a cylinder:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$2\pi r(r+h)$$

Curved area of a cylinder:

$$\pi r^2$$

$$2\pi rh$$

$$2\pi r^2$$

$$2\pi r(r+h)$$

1 m³ in litres:

100

1000

10 000

1 000 000

A pipe's cross-section area:

$\pi(R-r)^2$

$\pi(R^2-r^2)$

πR^2

$2\pi R$

5 cm to the nearest cm has bounds:

4–6

4.5–5.5

4.9–5.1

4.95–5.05

Cavalieri's principle compares:

bases

slice areas at every height

perimeters

diagonals

04

Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Concept map	Tushunchalar xaritasi	Карта понятий
Surface area	Sirt yuzasi	Площадь поверхности
Volume	Hajm	Объём
Bounds	Chegaralar	Границы
Compound solid	Murakkab jism	Составное тело
Self-assessment	O‘z-o‘zini baholash	Самооценка
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q4 asks for the curved area because:

it is easier

the question describes an open shape

it is always curved

no reason

A conversion mark is lost by:

using cm

mixing m and cm in one formula

using π

rounding

The volume bounds of a cuboid come from:

the stated values

all three extremes together

the largest edge

the average

Quarter III begins with:

- vectors
- the pyramid
- probability
- the cylinder

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Space diagonal of a $4 \times 6 \times 12$ cuboid

ANSWER 14

2. Surface area of a cube of edge 5

ANSWER 150

3. Volume of a prism, $B = 20$, $h = 9$

ANSWER 180

4. Volume of a cylinder, $r = 3$, $h = 8$

ANSWER 72π

5. 1 m^3 in litres

ANSWER 1000

6. Bounds of 7 cm to the nearest cm

ANSWER 6.5 and 7.5

7. Curved area, $r = 4$, $h = 10$

ANSWER 80π

Medium · the standard question

7 PROBLEMS

1. Angle of the $4 \times 6 \times 12$ space diagonal with the base

ANSWER $\approx 59.0^\circ$

2. Regular hexagonal prism, edge 5, height 11: total area

ANSWER $75\sqrt{3} + 330 \approx 460$

3. Trough 30/50 by 20 cm, 2.5 m long: litres

ANSWER 200

4. Cylinder holding 800 ml, height 10: radius

ANSWER ≈ 5.05 cm

5. Same: curved surface area

ANSWER ≈ 317 cm²

6. Pipe 2 m, $R = 9$, $r = 7$: metal volume

ANSWER $6400\pi \approx 20\,100$ cm³

7. Cuboid $5 \times 7 \times 9$ (nearest cm): volume bounds

ANSWER 248.6 to 391.9

Hard · stretch and reason

7 PROBLEMS

1. Same pipe: mass at 7.8 g/cm^3

ANSWER $\approx 157 \text{ kg}$

2. A silo, cylinder $r = 4$, $h = 12 \text{ m}$ with a hemisphere: volume

ANSWER $192\pi + \frac{128\pi}{3} \approx 737 \text{ m}^3$

3. A cylinder inscribed in a cube of edge 12: the ratio of volumes

ANSWER $\pi : 4$

4. A regular hexagonal prism of volume $540\sqrt{3}$ and edge 6: its height

ANSWER 10

5. A cuboid $3 \times 4 \times 5 \text{ m}$ to the nearest 0.1 m: volume bounds

ANSWER ≈ 58.4 to $\approx 61.7 \text{ m}^3$

6. Two similar prisms with volumes 8 and 125: the area ratio

ANSWER 4 : 25

7. A closed cylinder of volume 2000 cm^3 with least surface area: find r and h

ANSWER $r \approx 6.83$, $h \approx 13.66$

07 Homework

Homework — 4 tasks

Bring the concept map to the first lesson of Quarter III.

1. Rewrite in full every control-work question that lost a mark.
2. Finish the concept map with all six links written as sentences.
3. A cylindrical tank of radius 1.5 m and height 4 m: find its capacity in litres and its curved surface area.

4. Write your target for Quarter III in one checkable sentence, and date it.